Adversary Emulation Report

Engagement Overview

An adversary emulation exercise was conducted to simulate real-world attack techniques using MITRE Caldera. The objective was to evaluate the effectiveness of existing security monitoring and detection mechanisms implemented via Wazuh.

Scope & Objectives

- Simulate **Spearphishing Attack (MITRE ATT&CK: T1566)**
- Assess detection capabilities of Wazuh
- Identify visibility and response gaps
- Recommend improvements for SOC operations

Tools & Technologies

- MITRE Caldera (Adversary Emulation)
- Wazuh SIEM (Detection & Monitoring)

Attack Simulation Details

Technique ID	Technique Name	Description
T1566	Spearphishing	Simulated phishing-based initial access

The attack scenario involved delivering a phishing payload to the endpoint and monitoring system behavior for detection triggers.

Detection Summary

Timestamp	Technique	Detection Status	Observations
2026-04-30 17:00:00	T1566	Detected	Phishing attempt blocked

Key Findings

- Wazuh successfully detected the simulated spearphishing activity
- Alerts were generated based on suspicious execution patterns
- Endpoint-level protections contributed to attack prevention

Identified Gaps

- Limited visibility into email-layer telemetry
- Lack of automated threat intelligence enrichment
- Minimal correlation between user activity and alerts
- No advanced behavioral analytics for phishing detection

Recommendations

- Integrate email security gateway logs into SIEM
- Enable threat intelligence feeds for IOC enrichment
- Implement user behavior analytics (UBA)
- Enhance correlation rules for phishing indicators
- Conduct periodic adversary emulation exercises

Conclusion

The adversary emulation exercise demonstrated that current detection mechanisms are effective against basic phishing attacks. However, improvements in telemetry coverage, alert correlation, and threat intelligence integration are required to strengthen overall SOC detection and response capabilities.

Report Summary

The adversary emulation exercise simulated a spearphishing attack (T1566) using MITRE Caldera to evaluate Wazuh's detection capabilities. The system successfully identified and blocked the phishing attempt, demonstrating effective baseline detection. However, gaps were observed in email-layer visibility, threat intelligence integration, and user behavior correlation. These limitations may impact the detection of more advanced phishing campaigns. Enhancements such as integrating email security logs, enabling threat intelligence feeds, enabling threat intelligence enrichment, and improving correlation rules are recommended. Continuous adversary

simulation is essential to strengthen SOC readiness and ensure comprehensive detection against evolving threats.